CS 215: Task 05: Due: 04-Sep; time as announced on Piazza. No extensions.

- Your answers should be in your handwriting. Try to be neat and clear. If an LLM can read your handwriting, we might be able to.
- If you used any source (person or thing) explicitly state it. This is an individual assignment.

Teaching-team notes.

- When we say, 'derive' or 'show', simply put 2-3 important steps in the derivation in the space provided. Please attach extra paper for your records and for you to scan in the upload.

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0. Please write the honor code in your handwriting, with the understanding that it is true.

1. X and Y are independent random variables with MGFs $M_X(t)$ and $M_Y(t)$. Derive the MGF of

$$W = aX + bY + c.$$

Hence, find $E[W]$ and $\text{Var}(W)$.

Teaching-team notes. Use only the independence property mentioned in the slides (Aug 24-Slide 13).

2. $X_1, X_2, X_3 \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$. Let $T = X_{(2)}$ denote their median, i.e., the middle value when $X_{(1)} \leq X_{(2)} \leq X_{(3)}$ are in order. What is the pdf of T?

3. Let

$$X \sim \text{Bin}(n, p), \qquad Y \sim \text{Bin}(m, p)$$

be independent.

(a) Show that the conditional distribution of X, given $X + Y = k$, (i) is hypergeometric, and (ii) identify its parameters.

Teaching-team notes. Notation: $Z \sim \text{Hypergeometric}(N, W, K)$ if Z is the number of successes in a sample of size W drawn without replacement from a population of N objects containing K successes.

(b) Explain in words, without further calculation, why the parameter p disappears after conditioning.

4. A random ordering is made of a population containing K successes among N objects. In that ordering, let X be the number of successes among the first n_1 positions and Y the number among the next n_2 positions, where $n_1 + n_2 \leq N$. Find $\text{Cov}(X, Y)$.

   (a) Explain, without any calculation, the sign of the covariance.

   (b) Find $E[X]$ and $E[Y]$.

   (c) Find $\text{Cov}(X, Y)$.

5. Let $X_1, X_2, \ldots, X_k$ be independent random variables such that

$$X_i \sim \text{Geom}(p_i), \qquad i = 1, \ldots, k,$$

where the geometric distribution is supported on $\{1, 2, 3, \ldots\}$, i.e.,

$$P(X_i = n) = (1 - p_i)^{n-1} p_i, \qquad 0 < p_i < 1.$$

Define

$$M = \min(X_1, X_2, \ldots, X_k).$$

(a) Determine the distribution of M and its parameter(s).

(b) Find the exact probability that variable j achieves the minimum, i.e.,

$$P(X_j = M).$$

6. Let $X \sim N(0, \sigma^2)$ and $Y = |X|$. Let $\Phi(x)$ and $\phi(x)$ be the CDF and PDF of a standard normal random variable (i.e., mean 0, variance 1).

   (a) Derive the CDF and PDF of Y in terms of $\Phi(x)$ and $\phi(x)$.

(b) Compute $E[Y]$.

7. In an election, each of 1000 voters independently votes for candidate A with probability 0.52. We want to determine the probability that A receives a strict (i.e. ties not permitted) majority of the votes.

   (a) Complete the one-liner `python` script to compute this probability.

```
from scipy.stats import ????
prob =  ????
print(f"{prob:.4f}")
```

   (b) Use the normal approximation, the table in the Appendix of the textbook, and appropriate approximations to hand-calculate the probability without using any electronic aid.

(c) Use the normal approximation and `python` to compute the probability. Write the one-liner for the code fragment.

(d) Are you able to obtain the probability correct to 2 decimal places? If so, what approximations did you use?

8. The annual salaries of physicians are normally distributed. It is known that (in some currency)

   - 25% earn less than 180,000, and
   - 30% earn more than 320,000.

(a) Find the mean (μ) and standard deviation (σ) of the salary distribution.

(b) Find the fraction earning between 260,000 and 300,000.